

Selection, Recombination, or a Fresh Solve? A Candidate-Free Control for Single-Pass Test-Time Aggregation

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Abstract

When every candidate is wrong, correct-candidate selection is unavailable, yet the aggregation call can still solve the problem afresh. A correct aggregate answer may therefore reflect recombination, fresh solving, or both. For efficient test-time reasoning, the relevant question is whether candidate context adds value beyond the additional generation pass. We introduce the missing candidate-free control under the same maximum output-token allowance and stratify by the number of correct candidates. Across AIME-2025 and HMMT-2025 with Qwen3-4B, candidate conditioning improves accuracy when multiple candidates are correct ($\Delta_{\text{cand}}(\text{c2+}) = +0.290$), lowers accuracy when every candidate is wrong ($\Delta_{\text{cand}}(\text{c0}) = -0.123$), and remains unresolved in the one-correct regime. The c2+ and c0 conclusions survive a conservative correction for the adaptive two-benchmark procedure. Under this counterfactual, the interpretation of all-wrong recovery reverses at this scale: conditioning on an all-wrong candidate pool lowers accuracy relative to a fresh solve. Original-format matching and placebo results characterize the failures descriptively but leave their mechanism unresolved. Within a separate structured intervention, explicit answer fields causally steer outputs toward their values; masking yields no measurable accuracy improvement, and equivalence with the original format was not established. The evidence is limited to one Qwen3-4B family, two mathematics benchmarks, first-answer-truncated candidate fragments, and single-pass prompted aggregation.

1 Introduction

In an all-wrong candidate pool, correct-candidate selection is unavailable, yet the aggregation call can still solve the problem afresh. A correct aggregate answer may therefore reflect recombination, fresh solving, or some mixture of the two; its net value depends on performance relative to a candidate-free solve. Prior work often frames correctness on all-wrong pools as *recovery* and compares the aggregator with candidate-dependent baselines: majority voting (Wang et al., 2023), best-of-N (Cobbe et al., 2021), and learned selectors (Toshniwal et al., 2025). Such baselines score zero when every candidate is wrong, so a win over them leaves candidate-added value unresolved. Generation allowances may also differ sharply: for example, a 32,768-token trained refiner against a base model evaluated at 4,096 tokens (Wang et al., 2025, Appendix D.2).

An all-wrong recovery rate therefore combines any contribution from candidate context with the additional generation pass. We measure that contribution with a candidate-free generation under the same maximum output-token allowance. The comparison asks whether reading the candidates improves on using the aggregation-stage generation for a fresh solve.

We introduce that control. We evaluate a **single** aggregation pass with five arms: AGGREGATE; a candidate-free NO-CANDIDATE solve; ANSWER-ONLY, which receives answers without reasoning; VOTING; and ORACLE. All generating arms share one maximum generation-token allowance. Following Wang et al. (2025), we stratify by the number of correct candidates. A planted-recombination validation first checks that the harness can

detect recombination; a pre-specified cross-benchmark decision rule supports the central result.

The candidate-free control reveals a regime-dependent reversal. When multiple candidates are correct, conditioning on them improves accuracy ($\Delta_{\text{cand}} = +0.290$, raw pooled $p = 0.004$). When every candidate is wrong, conditioning on the same type of context reduces accuracy relative to a fresh solve ($\Delta_{\text{cand}} = -0.123$, raw pooled $p = 0.012$, Holm-adjusted 0.0236 under a retrospective conservative correction for the adaptive extension). In this all-wrong regime, the model’s wrong answers match displayed candidate answers far above coincidence (on AIME, 0.845 vs 0.371, $p < 0.0001$). The intermediate c1 regime remains unresolved. Both the deficit and the matching pattern persist in a pre-registered process-level audit rerun using fresh generations from the same process; under final-answer scoring, the c0 deficit *strengthens* (appendix H). A length-matched placebo control also shows descriptively that the matching is *relevance-insensitive*: off-topic candidates lifted from other problems reproduce much of the deficit (appendix I). In the redundant regime, where aggregation looks strongest, plurality voting scores numerically higher than aggregation (0.960 vs 0.887), although the paired difference is not statistically significant. The apparent recovery regime is therefore not where single-pass aggregation adds net value in this setting. A prompted, untrained $\leq 4\text{B}$ aggregator is in the regime where GSR’s scaling account predicts little refinement ability (§4; Wang et al., 2025); whether a net recombination benefit appears with training, scale, or iteration (§5) remains an open question.

Contributions. (i) We introduce the missing candidate-free counterfactual for measuring the conditional net value of candidate conditioning. (ii) We find a sign reversal across candidate-correctness regimes: conditioning helps when multiple candidates are correct and hurts when all are wrong; the c2+ and c0 conclusions remain significant under the conservative correction for the adaptive two-benchmark procedure. (iii) We show descriptively that all-wrong failures frequently match displayed answers and that off-topic candidate contexts reproduce much of the deficit. (iv) Within the structured intervention, explicit answer fields causally steer outputs, while masking those values does not measurably recover accuracy.

2 Method

Five arms, with a shared maximum generation-token allowance for the generating arms.

For each problem we draw a pool of independent candidate solutions and form sets of K candidates ($K = 4$ throughout, with a $K \in \{2, 4, 8\}$ sweep on AIME-2025 only). Each set is run through five arms. All generating arms share one maximum generation-token allowance (`max_new_tokens`). **AGGREGATE** sees the problem and K candidate solutions. Each candidate is truncated at its first valid answer, so the c0 finding of no net recombination benefit is scoped to first-answer-truncated solution fragments. The extraction argument in appendix A concerns the candidates’ *labels*, not their content. **NO-CANDIDATE** solves afresh from the problem alone. This candidate-free control measures the net effect of candidate conditioning relative to a candidate-free generation under the same maximum output-token allowance. **ANSWER-ONLY** sees the candidates’ final answers without reasoning. **VOTING** uses raw answer strings with a deterministic lexicographic tie-break, so equivalent surface forms may split votes. **ORACLE** marks a set solvable if any candidate is correct. Let C be the number of correct candidates and Y_a the final-answer correctness indicator under arm a . We define the conditional net effect of candidate conditioning once as

$$\Delta_{\text{cand}}(c) = \mathbb{E}[Y_{\text{AGGREGATE}} - Y_{\text{NO-CANDIDATE}} \mid C = c].$$

We use $\Delta_{\text{reas}} = \text{AGG} - \text{ANSWER-ONLY}$ as a descriptive contrast. All arms decode at temperature 1.0 (`top_p` 1.0, unseeded). All generating arms use the same 16,384-token maximum output allowance and decoding parameters. Realized prompt lengths, output lengths, latency, and total computation differ, so the comparison is neither compute-matched nor response-length-matched. At c0, where selection is unavailable, **AGGREGATE** – **NO-CANDIDATE** measures the net effect of conditioning on all-wrong candidates relative to a fresh generation under the same output cap. A positive contrast would be consistent with a net recombination benefit; a negative contrast shows that any recombination benefit is out-

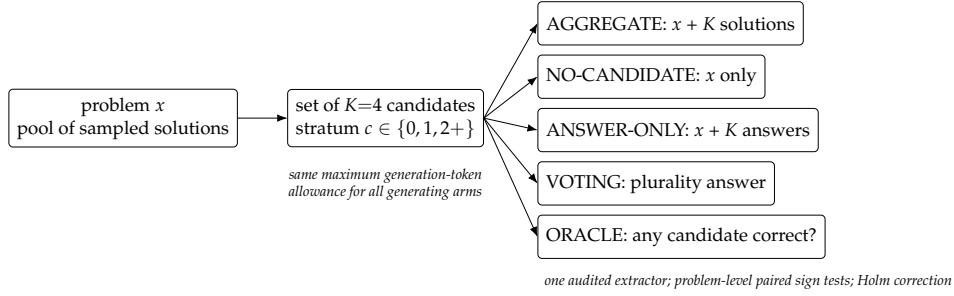


Figure 1: Evaluation design. Each set of $K = 4$ candidates is assigned to a stratum by its number of correct candidates and run through five arms; all generating arms share one maximum generation-token allowance. $\Delta_{\text{cand}} = \text{AGGREGATE} - \text{NO-CANDIDATE}$ measures the net effect of candidate conditioning relative to a candidate-free generation; selection is available only when $c \geq 1$; a surplus at $c = 0$ would be consistent with a net recombination benefit.

weighed by other effects of candidate conditioning. It does not establish that recombination never occurs.

Stratification. We bin each set by c , the number of its K candidates with a correct final answer: **c0** (none correct), **c1** (exactly one), and **c2+** (two or more). At **c0**, selection is unavailable. The $\text{AGGREGATE} - \text{NO-CANDIDATE}$ contrast therefore measures the net value of candidate conditioning in a selection-impossible regime. A positive contrast would be consistent with a net recombination benefit; a negative contrast establishes no net benefit in **c0**, not the absence of recombination. The c -bins are *constructed, conditional strata* — sets are drawn by conditioning on how many candidates are correct, not by sampling any deployment distribution — so every per-bin Δ is a conditional effect given the regime, and we make no claim about the deployment-weighted average of the three regimes (appendix C).

Labeling and analysis. One audited first-answer extractor strips `<think>`, truncates at the first valid answer, and matches gold; the pool’s stored correct is never used, so all arms are scored alike (first-answer labeling is required because traces run on past the answer; appendix A). Because a uniform extractor can still have an arm-specific *effect* — the arms’ text distributions differ by construction — the sensitivity of every headline number to the extraction rule is measured directly by a pre-registered dual-scoring sensitivity analysis (appendix H). The primary analysis is problem-level and clustered (one observation per problem per bin; exact two-sided sign test across problems; Holm within each Δ family of three bins). This is conservative, since set-level tests inflate power by treating repeated draws as independent, and it makes 30 problems per benchmark the binding constraint. Models: Qwen3-4B-Instruct-2507 and same-family RLVR Qwen3-4B-Thinking (Qwen Team, 2025), on AIME-2025 and HMMT-2025 (30 problems each). The primary Qwen3-4B-Instruct runs use a 16,384-token maximum output allowance; the exploratory Qwen3-4B-Thinking variant uses 32,768 tokens. Figure 1 summarizes the arms and the stratification.

3 Results

3.1 Candidate conditioning helps with multiple correct candidates and hurts when none are correct

The net effect of candidate conditioning changes sign with the availability of correct candidates. The three regimes appear in $\Delta_{\text{cand}}(c)$ (Table 1, pooled, problem-level): net-positive at **c2+** (+0.290, raw pooled $p = 0.004$) and net-negative at **c0** (−0.123, raw pooled $p = 0.012$), with **c1** directional (+0.242, $p = 0.076$). Because the second benchmark was collected adaptively, we pair the pooled tests with a retrospective conservative two-stage correction that charges for the extension: **c2+** and **c0** are rejected (Holm-adjusted $p = 0.0133$ and 0.0236

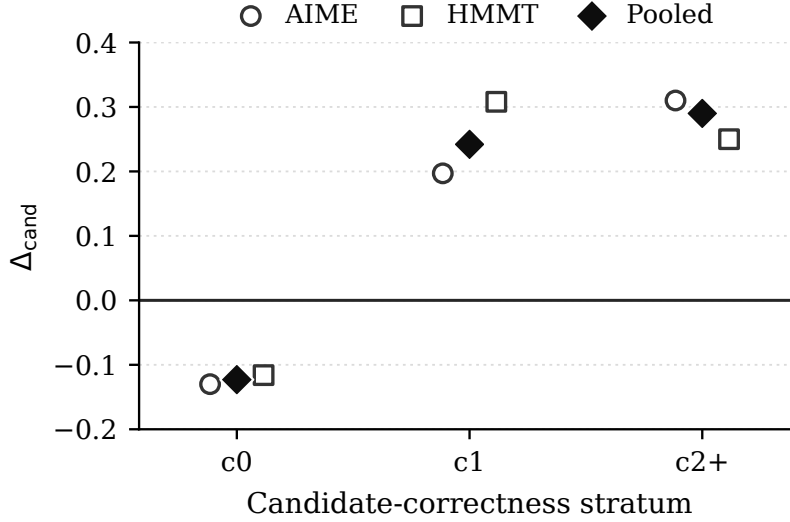


Figure 2: Candidate-conditioning effect by candidate-correctness stratum. Δ_{cand} is AGGREGATE minus NO-CANDIDATE. AIME, HMMT, and pooled estimates are shown. Markers show point estimates only; inferential results and the adaptive correction are reported in Table 1. Estimates are negative at c0 and positive at c2+; c1 is unresolved. The strata are constructed using gold correctness and do not represent deployment frequencies.

Table 1: Pooled problem-level accuracies by stratum (AIME+HMMT). $\Delta_{\text{cand}} = \text{AGG} - \text{NoCand}$; $\Delta_{\text{reas}} = \text{AGG} - \text{AnsOnly}$. The conservative adaptive correction rejects c2+ (adjusted $p = 0.0133$) and narrowly rejects c0 (0.0236 vs family $\alpha = 0.025$); c1 is not rejected (0.0755; Appendix L). [†]The c2+ Δ_{reas} estimate is descriptive because it does not persist under final-answer scoring (App. H). Bottom rows show the rerun estimates (App. H).

c	AGG	NoCand	AnsOnly	Vote	Oracle	Δ_{cand} (p)	Δ_{reas} (p)
c0	0.155	0.277	0.186	0.000	0.000	−0.123 (.012)	−0.032 (.359)
c1	0.672	0.430	0.539	0.156	1.000	+0.242 (.076)	+0.133 (.041)
c2+	0.887	0.597	0.710	0.960	1.000	+0.290 (.004)	+0.177 (.013) [†]
<i>fresh process-level rerun, final-answer scoring (App. H):</i>							
c0						−0.159 (< .001)	−0.073 (.115)
c2+						+0.242 (< .001)	+0.008 (1.00)

at family $\alpha = 0.025$, the latter narrowly), c1 is not (0.0755); the originally reported Holm correction at $\alpha = 0.05$ gives the same rejections (Appendix L). The sign reversal replaces a single “recovery rate” with a conditional account of when candidate context helps and when it harms; the c-bins are constructed conditional strata (§2) and do not imply deployment frequencies.

Planted-recombination validation. On constructed problems whose answer is recoverable only by combining two candidates that each carry half (all c0), AGGREGATE solves 0.767 of these planted additive-recombination tasks versus 0.000 for NO-CANDIDATE and ANSWER-ONLY (pre-specified success threshold 0.30; appendix F). In this construction, each successful recombination directly raises AGGREGATE accuracy, validating that the harness can express one additive form. For the real-data c0 sets, however, the contrast estimates the net effect of candidate conditioning, leaving the occurrence and rate of recombination unidentified.

At c2+, plurality voting scores numerically higher than aggregation (0.960 vs 0.887; Table 1, pooled); the paired difference is not statistically significant (10/4, $p = 0.180$).

The c_0 effect is exploratory on AIME (problem-level $p = 0.109$; set-level 0.016 discounted); that value fell in the pre-registered extension band (0.05, 0.15], which triggered the HMMT collection, and the HMMT stratum agrees in sign (-0.116). The pooled test is therefore adaptive by construction, and we never read the raw pooled p as if the second benchmark had been collected independently; the corrected analysis carries the inferential claim (Appendix L). The cross-benchmark agreement reduces concern that the pooled result is driven by one benchmark. We did not detect benchmark heterogeneity, and the benchmark-specific effects agree in sign; we therefore report the pre-specified pooled estimate while retaining the per-benchmark results (Table 5). The K -sweep, run on AIME-2025 only, is consistent: raw AGGREGATE accuracy at c_0 is flat from $K = 2$ to 4 (0.250, 0.222) then collapses at $K = 8$ (0.037), sign-invariant throughout (Table 6). The collapse is consistent with displayed-answer matching: more wrong candidates in context (eight vs two) means more wrong-answer strings available to match. These effects rest on ≈ 20 informative c_0 problems (further caveats, §5). A same-family reasoning-trained variant nearly empties the informative strata and is reported as exploratory in Appendix P. The c_0 deficit is also robust to the scoring rule: in the pre-registered process-level audit rerun under final-answer scoring it stays negative, gets *larger* ($-0.118 \rightarrow -0.159$), and remains significant after the same Holm procedure, including with truncated rows excluded (appendix H). Of the Table 1 entries, one is revised on that analysis: the $c_2 + \Delta_{\text{reas}}$ increment ($+0.177$) does not persist under final-answer scoring and is reported descriptively only (marked †; appendix H).

3.2 At c_0 , wrong outputs match displayed answers, even off-topic ones

The c_0 failures take a specific form. On AIME, when AGGREGATE errs at c_0 , its wrong answer matches a displayed candidate answer at 0.845, against a 0.371 coincidence rate for the candidate-free control; HMMT shows the same ordering (0.892 versus 0.506). ANSWER-ONLY sits between on both benchmarks (0.592 and 0.767; Figure 3 in the appendix), and the paired comparison on AIME is decisive (33/4, $p < 0.0001$; the p -value is AIME-specific, not pooled). We do not claim the matching *causes* the deficit: the per-problem matching-deficit correlation is null ($\rho = -0.23$, n.s.). Matching describes the form of the failures; the difficulty-stratified analysis shows that the accuracy loss is concentrated on otherwise-solvable problems ($\text{pass}@24 \geq 1$: $\Delta = -0.194$ AIME, -0.288 HMMT; near zero on problems no rollout solves; Table 10).

The matching pattern persists with off-topic candidate contexts. A placebo arm replaces the K candidates with length-matched solutions to other problems from the same pool — real reasoning, wrong topic — and reproduces much of the deficit (PLACEBO – NO-CANDIDATE = -0.164 , 95% percentile CI [-0.241 , -0.091]). The additional difference between genuine and off-topic candidates is small and its confidence interval includes zero (AGGREGATE – PLACEBO = $+0.045$, [-0.018 , $+0.118$]). More than half of the wrong placebo outputs — 104 of 192 — matched an off-topic answer shown in the prompt, showing that displayed-answer matching remains substantial even when candidate content is unrelated to the target problem. The all-wrong deficit does not require topically relevant candidate solutions. The experiment does not uniquely identify whether the harm arises from long-context degradation, task reframing, displayed-value steering, or their interaction. These results characterize the original-prompt failures descriptively; the structured intervention tests one candidate-answer channel causally (§3.3, appendix I).

3.3 A structured answer-field intervention

To test one displayed-answer channel, we re-rendered all 220 eligible all-wrong (c_0) sets from 55 problems across both benchmarks under three conditions: FULL, MASKED, and SUBSTITUTED. Each condition retained byte-identical reasoning prefixes and changed only the explicit answer field. FULL preserved the candidate’s original answer, MASKED replaced it with [ANSWER MASKED], and SUBSTITUTED supplied a controlled wrong value. Across the two Holm-corrected confirmatory endpoints, with problem-clustered 95% percentile CIs, endpoint B measured value-specific steering: $P(\text{a SUBSTITUTED output matches an injected value}) - P(\text{a FULL output matches the same values}) = +0.0682$, 95% CI [$+0.0227$, $+0.1182$], Holm-adjusted $p = 0.0425$. Endpoint A measured accuracy recovery from mask-

ing: $\text{acc}(\text{MASKED}) - \text{acc}(\text{FULL}) = -0.0045 [-0.0500, +0.0364]$, Holm-adjusted $p = 1.000$. Thus, within the structured intervention, changing explicit answer fields causally steers outputs toward their values, while masking those fields yields no measurable accuracy improvement. Appendix M reports the figure and sensitivity analyses.

Because the structured rendering differs from the original prompt, we also evaluated a bridge on the same 220 sets. FULL-structured minus original-prompt AGGREGATE was $+0.0227$, with a 95% CI of $[-0.0182, +0.0682]$, against a pre-fixed equivalence margin of ± 0.05 . Equivalence with the original prompt was not established. The causal result is therefore confined to the structured intervention; the original-format matching and placebo results remain descriptive, and the mechanism behind the original deficit remains unresolved.

4 Related work

Prior work on aggregation has emphasized whether a method can recover a correct answer when correct-candidate selection is unavailable. GSR (Wang et al., 2025) generates a candidate pool, synthesizes a refined answer, and stratifies by the number of correct candidates. On all-wrong pools, where majority voting and best-of-N must fail, its refiner answers a small fraction correctly ($\leq 9\%$ at $N_c=0$ across its benchmarks; Table 5), interpreted as recombination. SSA (Qi et al., 2025) trains an aggregator that outperforms majority voting ($+8\%$ pass@5 MATH) and a 72B reward-model re-ranker. GenSelect (Toshniwal et al., 2025) instead reasons to select the best of N and is bounded by pass@N; its follow-up (Toshniwal et al., 2026) filters to pools with ≥ 1 correct candidate, where selection remains available. Together with self-consistency (Wang et al., 2023) and best-of-N (Cobbe et al., 2021), these methods span selection and synthesis from candidate pools.

Our question is complementary: does reading those candidates improve the answer relative to using the aggregation-stage generation for a fresh solve? None of these studies includes a candidate-free generation under the aggregation stage’s allowance within candidate-correctness strata. GSR reports call-matched majority-voting and base-model comparisons. GenSelect’s Table 2 compares allocating an inference budget entirely to more generations followed by majority voting with splitting it between generation and selection. Our design instead holds the candidate-correctness stratum fixed and compares candidate conditioning with a candidate-free generation, including the all-wrong regime where selection is unavailable.

GSR and SSA are single-pass methods. Iterative methods such as RSA (Venkatraman et al., 2025) and TRT (reaching 100% on AIME by iterating; Zhuang et al., 2026) form a distinct multi-round class outside our scope (§5). RSA gives its majority-voting baseline the same total generation allowance. Its Appendix-F example, using our exact model, recombines correct intermediate steps from imperfect candidates and adds structure absent from any individual candidate (appendix D), demonstrating that the mechanism can occur. The conditional net effect remains a separate quantity, for which RSA does not include a candidate-free arm.

The distinction changes the interpretation of all-wrong recovery. In our setting, the aggregator has a positive raw recovery rate on all-wrong pools, but its candidate-free contrast is negative ($\Delta_{\text{cand}} = -0.123$). Successful answers in this regime therefore do not by themselves show that candidate conditioning improved accuracy. This result is compatible with GSR: its untrained prompted aggregator also has a near-zero refinement gap, the regime our study occupies (appendix B). Whether training or iteration produces a positive net recombination benefit remains open. We credit GSR for the candidate-correctness stratification; our contribution is the candidate-free control and the reversed all-wrong conclusion for single-pass aggregation.

5 Limitations

The study covers one Qwen3-4B model family at one principal size, one deterministic candidate-set construction seed (with unseeded model decoding), and two 30-problem

mathematics benchmarks. The c0 effect is significant only in the pooled analysis, and its corrected margin is narrow (Holm-adjusted 0.0236 at family $\alpha = 0.025$). Candidates shown to the aggregator are truncated at each one’s first valid answer, so the c0 finding of no net recombination benefit is established only for first-answer-truncated fragments, not full traces. AIME-2025 and HMMT-Feb-2025 may be affected by benchmark contamination. This may inflate absolute accuracy; paired within-problem contrasts are less directly affected but are not immune. The original-prompt mechanism evidence (matching, placebo) is descriptive. The causal result is scoped to the structured intervention. Equivalence between the structured and original prompts was not established, so the two forms of evidence cannot be conflated. On a prospectively frozen 30-set timing sample, pipeline-rendered, cold-prefix, batch-size-1 median request latency was 13.9/13.0/14.3 seconds for AGGREGATE versus 55.8/59.4/38.8 seconds for a fresh solve in c0/c1/c2+, respectively. AGGREGATE read substantially longer prompts but generated much shorter outputs. These descriptive marginal post-pool measurements disabled prefix caching and excluded candidate generation; they are neither cached-serving nor end-to-end cost estimates (Appendix N, Table 3). The extraction-scoring sensitivity analysis (appendix H) shows the c0 deficit is robust to the scoring rule; the c2+ advantage over ANSWER-ONLY is reported descriptively only, as it does not persist under the final-answer rule. The observed matching and steering behavior may also depend on the tested candidate rendering. The canonical AGGREGATE prompt displays each candidate with an explicit “[Candidate *i*] (final answer: *X*)” line before its retained solution text, making value-salience effects format-dependent. These format-specific findings may not extend to prompts that omit displayed answer strings. The structured intervention is additionally scoped to its “Reported final answer” rendering. No non-Qwen or non-mathematics replication exists yet; a cross-family check (e.g., DeepSeek-class models, Guo et al., 2025) is a designed next step.

6 Conclusion

An all-wrong pool removes correct-candidate selection, yet an aggregator may still answer correctly by solving the problem afresh. The candidate-free counterfactual reveals a regime reversal: candidate context helps when multiple candidates are correct, harms when every candidate is wrong, and leaves the one-correct regime unresolved. The c0 contrast estimates the net effect of candidate conditioning, leaving the occurrence and rate of recombination unidentified. Original-format matching and placebo results characterize the failures descriptively. Within the structured intervention, explicit answer fields causally steer outputs, whereas masking yields no measurable accuracy gain; the bridge analysis did not establish equivalence with the original format. These conclusions are limited to one Qwen3-4B family, two mathematics benchmarks, first-answer-truncated candidates, and single-pass prompted aggregation. For efficient-reasoning evaluation, candidate-added accuracy, marginal post-pool latency, and end-to-end cost are distinct quantities and should be reported separately. Evaluations that interpret correctness on all-wrong candidate pools as evidence of aggregation benefit should include a candidate-free generation control at the aggregation stage.

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A Answer extraction and verifier audit

Correctness is determined by a single audited first-answer extractor rather than the pool’s stored label; an audit against the benchmark golds found no false negatives. Instruct traces continue past their first answer (run-on rate 99.3%), so a last-answer label would disagree on $\sim 20.8\%$ of traces. We apply the first valid answer uniformly across arms. Any residual labeling error would attenuate the headline result: a truly-c1 set misclassified as c0 gives AGGREGATE a selectable correct answer, raising apparent c0 accuracy and weakening the matching estimate.

The run-on and disagreement figures above (99.3%, $\sim 20.8\%$) are statistics of the candidate *pool*, so this appendix supports the pool’s c-bin **labels**. Because candidates shown to

AGGREGATE are truncated at each candidate’s first valid answer (§2), the c0 finding of no net recombination benefit is scoped to first-answer-truncated solution fragments; full traces lie outside this scope. Arm-output narration is assessed separately in the pre-registered dual-scoring audit of appendix H, which uses fresh generations from the same process and evaluates first- and final-answer scoring side by side.

B Extended comparison to GSR, SSA, and TRT

GSR reports a compatible scaling pattern: its Figure 2 (“A comprehensive study of refinement gap”) shows the gap scaling with model size and “weakly correlated with the base capability,” while its prompted, untrained aggregator shows essentially no gap over majority voting (Table 1: untrained selfRef@4 average Δ of ± 0.0). These results place our null alongside GSR’s scaling account. GSR’s deployed prompt explicitly instructs the model *“Don’t copy candidates, use insights selectively and reason independently,”* whereas displayed-answer matching remains present with our separately designed prompt. Because both the prompts and studies differ, a controlled prompt intervention would be needed to determine whether alternative wording mitigates the effect.

SSA’s error analysis (§A.6 and Table 9) reports that the vast majority of correct outputs are *copied* from a candidate that already held the right answer. The rare correct-yet-gold-absent cases are mostly wrong-format outputs missed by extraction, leaving little evidence of genuine synthesis in that subset. These results are consistent with selection-dominant gains in a trained aggregator. Their §A.7 answer-truncation probe nevertheless argues that some synthesis ability remains, limiting how strongly the copy-dominance result should be read. TRT’s loop (generate $\rightarrow$ self-rank $\rightarrow$ select-best $\rightarrow$ update strategy) combines diversity with self-verification while leaving recombination separately unidentified. On code, it also executes self-generated tests, so its gains incorporate an execution signal in addition to model self-assessment.

C What the regimes would imply if observable

If the hidden regime were known, the observed point estimates numerically favor plurality voting at c2+, while a fresh solve scores higher at c0; the c1 comparison remains unresolved. This remains a conditional observation rather than a deployment recommendation. The regime is unavailable in practice because c is defined by correctness against gold, and the c-bins are constructed strata (§2). The analysis therefore supplies neither real-workload regime frequencies nor a deployment-weighted value of aggregation. Within these strata, the c0 deficit occurs on problems the model could otherwise solve. The placebo control (appendix I) finds much of the same deficit with off-topic solutions, leaving the benefit of relevance filtering unresolved.

D RSA: baselines and the Appendix-F recombination example

RSA’s headline (its Table 1; K=4, N=16, T=10 steps, 4 seeds) budget-matches its parallel baselines: on AIME-2025, Pass@1 = 73.18 ± 2.20 for RSA versus 68.33 for budget-matched majority voting and 43.91 for the base model; on HMMT-2025, 47.55 versus 35.00 versus 27.17. Per-round accuracy appears only as a figure (Pass@1 improves monotonically with RSA steps, their Fig. 6), so we cite the trend qualitatively and no round-level number. RSA’s headline comparison already controls the total generation allowance across its parallel baselines. The additional counterfactual required for our question is a candidate-free arm. Its Appendix-F case study (Qwen3-4B-Instruct-2507 — our exact model — on “the sum of the positive divisors (including 1) of 9! that have units digit 1,” answer 103) draws correct intermediate steps from candidates with imperfect or partial traces and produces a solution whose structure is absent from every individual candidate. This provides a clear qualitative instance of recombination-plus-addition in an iterative setting. Our separate result is that, when every candidate is wrong, candidate conditioning has a negative net effect relative to a fresh solve in the single-pass setting tested here.

E Pre-specification and analysis plan

The analysis plan was fixed before the confirmatory (HMMT) data were collected. The primary analysis is problem-level and clustered: each problem contributes one observation per c-bin (its mean accuracy per arm across its sets), followed by an exact two-sided sign test across problems and a Holm correction within each Δ family (three c-bins). Under the pre-committed AIME→HMMT rule, an AIME-only problem-level c0 Δ_{cand} sign-test $p < 0.05$ would be reported directly; a p-value in the gray zone $(0.05, 0.15]$ would trigger one pre-registered extension to new problem units from HMMT, after the verifier audit passed on HMMT golds; and $p > 0.15$ would be reported at set level only. The observed AIME problem-level $p = 0.109$ fell in the gray zone, so the single planned extension was conducted. Pooling then gave c0 $\Delta_{\text{cand}} = -0.123$, with raw pooled $p = 0.0118$ and Holm-adjusted $p = 0.0236$ under the conservative stage-2 family level $\alpha = 0.025$.

Two observations support this chronology. (1) Cross-benchmark sign agreement is informative: selective tuning to AIME alone offers no reason to expect the same direction on a second disjoint benchmark. (2) A commit predating the HMMT run also records the plan, although we treat it as corroborating evidence because the OpenTimestamps anchor post-dates the run and establishes tamper-evidence only prospectively. The *negative* c0 direction itself was exploratory on AIME and then confirmatory on HMMT; it was not pre-registered as a directional prediction.

F Construction of the validation experiment

The validation benchmark is a constructed-recombination benchmark run through the identical harness: each problem’s answer is $X+Y$, one candidate carries X and one carries Y , both report wrong final answers, and every set is c0, so the answer is recoverable only by composing the two candidates. The pre-specified success criterion requires c0 AGGREGATE to exceed both NO-CANDIDATE and ANSWER-ONLY by ≥ 0.30 . Observed: AGGREGATE 0.767 vs 0.000/0.000. The validation establishes that the harness can express and detect this constructed additive form of recombination. Its scope is specific to that mechanism: sensitivity to heterogeneous natural recombination remains uncalibrated, and the observed success supplies no lower bound on their rates.

G K-sweep and run-health details

On AIME-2025, Δ_{cand} at c0 is sign-invariant across $K \in \{2,4,8\}$, with raw AGGREGATE accuracy decreasing from 0.250 to 0.222 to 0.037; the per-K table is Table 6 and the run-health table is Table 9. The instruct and HMMT runs have truncated-no-answer and candidate-starvation counts of 0 on every arm. The only nonzero run-health counts occur for RLVR: 37/240 NO-CANDIDATE and 13/240 ANSWER-ONLY truncations, versus 0 for AGGREGATE. These counts deflate the comparison arms and bias the RLVR c2+ Δ_{cand} *upward* (in favor), so we treat it as an exploratory upper bound.

Truncation (hitting the token cap, with or without an answer) is *arm-asymmetric by construction*: the candidate-free arms spend the budget reasoning from scratch while AGGREGATE mostly reads. In the process-level rerun (appendix H), the instruct truncation rates are NO-CANDIDATE 41/472 (8.7%) and ANSWER-ONLY 20/472 (4.2%), versus AGGREGATE 3/472 (0.6%). All arms nevertheless have **zero** truncations without an extractable answer (trunc_noans 0/472), and first-answer scoring is unaffected by post-answer run-on. Truncation can influence the *final-answer* rule, so the rerun reports every flip statistic by truncation status and bases its primary interpretation on non-truncated rows (appendix H). All tables regenerate from the committed logs via `make_numbers.sh` and `make_display_items.py`.

H Extraction-scoring sensitivity analysis

Potential extraction asymmetry (pre-registered). Arm outputs are scored by a first-valid-answer extractor (§2, appendix A). AGGREGATE and ANSWER-ONLY see candidate answers, and the AGGREGATE prompt highlights each candidate’s final answer on its own line. A generation may therefore mention a candidate answer before giving its own commitment (“Candidate 1’s answer is 50, let me check... final answer 42”), causing the first-answer extractor to score 50. NO-CANDIDATE has no candidate answers in its prompt. At c0, where all candidates are wrong, and c2+, where multiple candidates are correct, this asymmetry could produce the observed c0 deficit, c2+ gain, and elevated matching rate through the scoring rule alone. We preregistered the design, endpoints, thresholds, and decision rule before generating the rerun data. The project archive retains the preregistration, analysis script, and machine-generated output. These rerun materials are not part of the workshop code supplement. Every number in this appendix and Appendix I comes from that output, while the primary pipeline and its locked numbers sheet remain unchanged.

Design. The original arm texts were not retained. The rerun therefore generates and logs fresh raw outputs over the full grid at the primary settings (1,416 generations across both benchmarks; identical pools, seed-0 set composition, budget, $T=1.0$). It provides a **process-level** estimate from new draws under the same generation process. The committed results remain the primary reference. Two rules score every logged text side by side: (1) the primary **first-answer** extractor, unchanged; (2) a **final-answer** extractor, `extract_final_answer_tiered`, with the documented hierarchy `last \boxed{} > <answer> tag > last “final answer:” cue > last line`, recording which tier fired. The primary statistic is the **asymmetry** flip(AGG) – flip(NoCand) at c0. NO-CANDIDATE provides the zero-exposure reference because its prompt contains no candidate answer. Flips are problem-clustered with 95% bootstrap CIs ($B=10,000$) and **stratified by truncation**. When a generation reaches the token cap before committing an answer, the final-answer rule falls through to `last_line`, producing a scoring disagreement driven by termination rather than candidate narration. Truncated rows are therefore reported separately and excluded from the primary claims. We inspected the full text of every disagreement using a retained manual-audit file, which is not part of the workshop code supplement.

The pre-registered §6 decision rule defines the asymmetry as “small” when its CI upper bound is < 0.05 and below the committed $|\Delta_{\text{cand}}| \approx 0.13$. It retains the c0 deficit when the estimate remains negative and significant under the same Holm procedure used for the accepted-paper headline on non-truncated rows. It retains the matching result when the AGG–NoCand matching asymmetry remains positive at $\geq 50\%$ of its first-answer magnitude. The same standard applies within the c2+ Δ_{cand} and c0/c2+ Δ_{reas} families; a criterion that is not met triggers restatement of the affected family. Interpretation also requires the first-answer rerun to reproduce the committed signs and Holm outcomes. Otherwise, the rerun is inconclusive for the primary results.

Run checks. All pre-specified checks passed: set reconstruction against the committed logs (0 mismatches on 472 sids), committed sid-map cross-check, `prompt_sha256` verified on all 1,636 rows, launch canary (no empty texts; worst within-batch duplicate-text rate 0.008), and row-level re-extraction checks (0/1,636 mismatches on text hash, extractor output, and correctness).

Results. The rerun met the replication criterion: all six per-benchmark Δ_{cand} signs match the committed run, and the same effects remain significant after Holm under first-answer scoring (c0 $\Delta_{\text{cand}} -0.118$; c2+ $\Delta_{\text{cand}} +0.290$). The observed flip asymmetry has the opposite sign from the proposed narration artifact:

arm (c0, all rows)	flip rate [95% CI]	answer-level disagreement
AGGREGATE	0.045 [0.009, 0.091]	0.132
NO-CANDIDATE	0.077 [0.041, 0.118]	0.377
ANSWER-ONLY	0.105 [0.059, 0.155]	0.377

The primary asymmetry $\text{flip}(\text{AGG}) - \text{flip}(\text{NoCand})$ is -0.032 $[-0.086, +0.027]$ (non-truncated rows: -0.053 $[-0.121, +0.015]$). Its CI upper bound, $+0.027$, is below the pre-registered 0.05 margin. AGGREGATE shows the smallest change under the two scoring rules, with $\geq 99\%$ of its final-answer extractions using the boxed tier. The pooled, problem-level results under both rules, with Holm correction within each family, are:

quantity	first-answer	final-answer	verdict
c0 Δ_{cand}	-0.118 ($p = 0.012$)	-0.159 ($p < 0.0001$)	survives, strengthens (Holm REJ, also excl. trunc.)
c0 Δ_{reas}	-0.041 ($p = 0.263$)	-0.073 ($p = 0.115$)	sign preserved
c2+ Δ_{cand}	$+0.290$ ($p = 0.0002$)	$+0.242$ ($p = 0.0003$)	survives (Holm REJ)
c2+ Δ_{reas}	$+0.161$ ($p = 0.0013$)	≈ 0 (Holm $p = 1.000$)	fails — restated
c0 matching AGG / NoCand	0.885 / 0.417	0.874 / 0.475	ordering holds ($\geq 50\%$ retention)

The committed per-benchmark rates in §3.2 (AIME: 0.845/0.371; HMMT: 0.892/0.506) come from the original run, whereas the values here are pooled process-rerun rates under the extraction audit. The absolute rates differ because the run and pooling differ; the ordering is unchanged.

Final-answer scoring produces a larger c0 deficit than first-answer scoring, while the matching attribution changes little. A match disappears for 15 of 161 AGGREGATE matching flags: manual review identifies 9 genuine self-corrections in which a narrated candidate answer precedes the correct boxed answer, and 6 revisions to a different wrong answer with no accuracy change. The c0 deficit is also present on **clean-c0**, the 178 sets whose candidates are wrong under *both* rules ($\Delta_{\text{cand}} -0.099$, $p = 0.013$; matching 0.877). The finding therefore remains when c-bin membership agrees under both scoring rules. The realized-token log supports the §2 budget description: at c0, AGGREGATE reads $\sim 18.7\text{k}$ prompt tokens and generates $\sim 1.7\text{k}$, while NO-CANDIDATE reads $\sim 0.2\text{k}$ and generates $\sim 8.3\text{k}$. The arms share a maximum output-token allowance, while their realized computation differs substantially.

Final-answer scoring changes the conclusion for the c2+ Δ_{reas} increment ($\dagger$ in T1). It is significant under first-answer scoring ($+0.177$ committed; $+0.161$ in the rerun, Holm REJ) and indistinguishable from zero under final-answer scoring. Manual review finds 24 ANSWER-ONLY c2+ rows that flip wrong $\rightarrow$ correct, all on the boxed tier. In each case, ANSWER-ONLY produced a correct boxed answer, while first-answer extraction selected an earlier intermediate value. In accordance with the pre-registered rule, the c2+ reasoning increment over ANSWER-ONLY is treated as unresolved. The central AGGREGATE-versus-NO-CANDIDATE contrast is unchanged.

I Long-context placebo control (PlaceboLong)

The extraction analysis addresses scoring sensitivity. Context length remains a separate potential explanation: AGGREGATE is a *long-context* arm (reading $\sim 6\text{--}8\text{k}$ candidate tokens), whereas NO-CANDIDATE and ANSWER-ONLY use short contexts. The c0 deficit could reflect candidate content, generic degradation from long noisy context, or both. A control arm pre-registered in the same commit and subject to the same vetting as appendix H (pre-reg §9a) examines this possibility. At c0 only, it replaces the K candidates with **real first-answer-truncated traces from other problems in the same pool**, matched by nearest solution word-length. Placebos whose answer is accidentally correct for the target problem are excluded, and a separate pre-registered RNG seed leaves the main experiment’s set composition bit-identical. The resulting context is length-matched and locally coherent, but concerns the wrong problem.

Problem-clustered c0 accuracies: AGGREGATE 0.173, PlaceboLong 0.127, NO-CANDIDATE 0.291. The two pre-registered gaps: the off-topic-context contrast (registered label **distraction**) = PlaceboLong – NoCand = -0.164 $[-0.241, -0.091]$ (CI excludes 0); the on-topic increment (registered label **content**) = AGG – PlaceboLong = $+0.045$ $[-0.018, +0.118]$ (CI includes 0). The registered labels name the two contrasts without imposing a unique causal

decomposition. Off-topic candidates reproduce much of the deficit, and the AGGREGATE-PLACEBO contrast is not statistically distinguishable from zero. Under the pre-stated classification rule, this result weighs against a content-specific account. More than half of the wrong placebo outputs, 104 of 192 (a row-level rate of 0.542), matched an *off-topic* answer shown in the prompt. Displayed-answer matching therefore remains substantial when the candidate content is unrelated to the target problem. We describe this pattern as **relevance-insensitive displayed-answer matching**. The deficit can occur without topically relevant candidates; the respective roles of long-context degradation, solution-shaped task reframing, displayed-value steering, and their interactions remain unresolved. This secondary mechanism check leaves the §3 primary statistics unchanged. The regime discussion in appendix C incorporates the result, while the matching-deficit correlation retains its descriptive interpretation (§3.2).

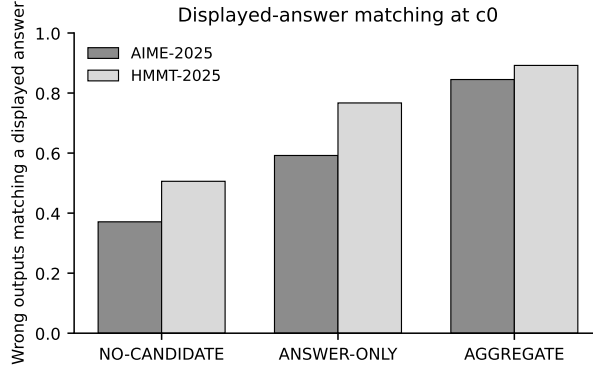


Figure 3: Wrong c0 outputs matching a displayed candidate answer, by arm (AIME-2025, HMMT-2025). The paired comparison (33/4, $p < 0.0001$) is AIME-specific. Matching means that the arm’s wrong answer equals one of the candidate answers displayed to AGGREGATE.

J Identification notes

Two observations clarify the identification limits. First, an algebraic identity links the placebo arm to the primary contrast:

$$\Delta_{\text{cand}}(0) = (\text{AGGREGATE} - \text{PLACEBO}) + (\text{PLACEBO} - \text{NO-CANDIDATE}).$$

The equality is algebraic. Interpreting the first term as a topical-content increment and the second as a generic-context effect requires a placebo-exchangeability assumption (that off-topic candidates differ from genuine ones only in topical relevance), and the mechanism components inside either term are not separately identified. Second, arm-level correctness contrasts do not identify a unique recombination rate: different latent mixtures of recombination, fresh reasoning, interference, and steering can produce the same arm means. This conclusion applies to arm-level correctness contrasts. The retained raw transcripts may support later text-level provenance analysis.

K Additional related work

The candidate-free comparison speaks to the broader test-time-scaling literature. Repeated sampling scales coverage predictably (Brown et al., 2024), and compute-optimal test-time strategies can beat parameter scaling (Snell et al., 2025). Our NO-CANDIDATE arm supplies the corresponding fresh-solve comparison, using another generation under the same maximum output-token allowance; realized prompt length, output length, latency, and computation differ across arms. Under an equal-total-cost accounting, Sharma & Chopra (2025) likewise find that sequential refinement can outperform parallel self-consistency.

Together, these results illustrate how the comparison and resource accounting can affect conclusions.

The self-improvement family — Self-Refine (Madaan et al., 2023), Reflexion (Shinn et al., 2023), Tree-of-Thoughts (Yao et al., 2023), and STaR (Zelikman et al., 2022) — asks the model to read and revise its own prior outputs. Our c0 result illustrates a related risk: in the tested setting, conditioning on wrong prior answers can reduce accuracy relative to a fresh solve. Process supervision (Lightman et al., 2024) and benchmark design (Hendrycks et al., 2021) shape what “a correct candidate” means in these pools; we score final answers against gold with an audited extractor (appendix A) and leave step-level credit to future work.

L Correcting for the adaptive extension

The two-benchmark design was adaptive. The analysis plan committed a collection rule in advance: extend to a second benchmark if and only if the first benchmark’s problem-level c0 p-value fell in $(0.05, 0.15]$. The observed AIME-2025 value, $p_1 = 0.109375$, triggered the extension, and the headline c0 test was then run on the pooled data. Because the extension decision depended on the first test, subsequently testing the pooled data created an additional opportunity for rejection beyond the originally reported Holm family. We therefore applied a *retrospective conservative* two-stage correction to the executed adaptive procedure. This correction was developed after the experiment.

At stage 1, the c0 effect may be claimed from the first benchmark alone only when its raw $p_1 \leq 0.025$. At stage 2, entry into the pre-committed extension band leaves the collection rule unchanged and tests the pooled family $\{c0, c1, c2+\}$ with Holm at family level $\alpha = 0.025$. The union bound gives $\text{FWER} \leq 0.05$ (0.025 stage-1 + 0.025 Holm stage-2) under any dependence and any partial-null configuration, without relying on simulation or asymptotics. We also re-derived the originally reported values from the committed per-problem logs using the same exact sign test as the rest of the paper (stage-1 c0: 0.109375 against the reported 0.109; pooled c0: 0.011818 against the reported 0.012).

The observed data follow the extension path. Holm at family $\alpha = 0.025$ gives the adjusted p-values in Table 2: c2+ and c0 are rejected, while c1 is not rejected. The originally reported Holm correction at $\alpha = 0.05$ gives the same two rejections with identical adjusted values. The c0 rejection is narrow (adjusted $p = 0.0236$ against 0.025). The original analysis did not claim statistical significance for c1, and c1 remains not statistically significant under the corrected analysis.

As supplementary calibration, we simulated the complete adaptive procedure (stage-1 test, extension band, pooled tests, Holm) under the global null, applying one Rademacher sign flip per problem, consistently across every stratum that problem populates ($B = 200,000$; 60 problems). The originally reported procedure’s empirical family-wise error under the global null is 0.0270 (Monte Carlo 2 s.e. ± 0.0007) when the raw c0 p governs the stage-1 claim, and 0.0079 under a stricter reading that additionally requires a stage-1 Holm pass; the conservative two-stage rule’s is 0.0228. This simulation provides a global-null calibration. Strong family-wise error control under partial-null configurations follows from the union-bound argument above.

Table 2: The adaptive two-benchmark extension, retrospectively corrected. Stage 1 (claim from the first benchmark alone) requires raw AIME c0 $p_1 \leq 0.025$; the observed $p_1 = 0.109375$ instead entered the pre-committed extension band, so the pooled family is tested with Holm at family $\alpha = 0.025$. $\text{FWER} \leq 0.05$ by union bound, valid under any dependence and any partial-null configuration. The originally reported Holm correction at $\alpha = 0.05$ yields the same rejections with identical adjusted values.

stratum	pooled p	Holm-adj. p	reject ($\alpha_{\text{fam}}=0.025$)	reject ($\alpha=0.05$)
c2+	0.004425	0.0133	yes	yes
c0	0.011818	0.0236	yes	yes
c1	0.075519	0.0755	no	no

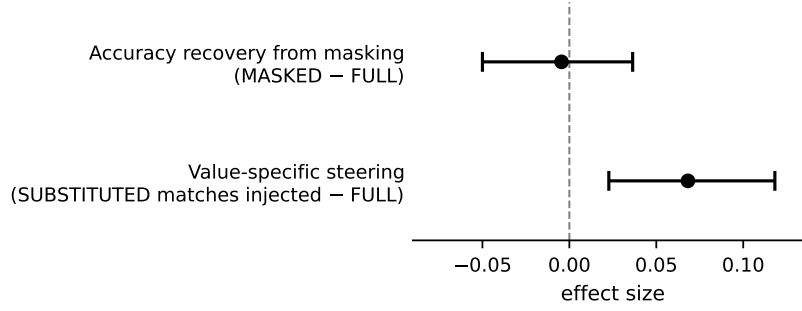


Figure 4: Structured answer-field intervention on all-wrong (c0) candidate sets — structured intervention only. Top: accuracy recovery from masking (MASKED - FULL), point -0.005 , 95% percentile CI $[-0.050, +0.036]$. Bottom: value-specific steering (SUBSTITUTED output matches an injected value minus FULL output matching the same values), point $+0.068$, 95% percentile CI $[+0.023, +0.118]$. Problem-clustered percentile bootstrap, 55 problems and 220 sets. Causal conclusions are scoped to the structured intervention.

M The structured answer-field intervention: details

All 220 eligible all-wrong sets were re-rendered with the candidate’s terminal answer commitment removed from its retained reasoning prefix and an explicit “Reported final answer” field added; the three conditions (FULL, MASKED with the field set to [ANSWER MASKED], SUBSTITUTED with a controlled wrong answer never equivalent to gold and agreement-preserving across equivalent candidates) share byte-identical prefixes, yielding 660 generations under the same maximum generation-token allowance with unseeded decoding. First-answer scoring is primary, with the audited final-answer rule as a pre-specified sensitivity; Holm is applied over the two confirmatory endpoints; all intervals are problem-clustered 95% percentile CIs. Full endpoint statistics: endpoint B $+0.0682$ $[+0.0227, +0.1182]$, sign-test $p = 0.0213$, sign-flip permutation $p = 0.0102$, Holm-adjusted 0.0425 (reject); endpoint A -0.0045 $[-0.0500, +0.0364]$, sign-test $p = 1.000$, Holm-adjusted 1.000 (do not reject). Sensitivities: B under the audited final-answer rule $+0.0636$ $[+0.0182, +0.1136]$, $p = 0.0352$; B under truncation exclusion $+0.0773$ $[+0.0273, +0.1303]$, $p = 0.0213$; A under the same two sensitivities $+0.0182$, $p = 0.344$ and -0.0091 , $p = 1.0$. The upstream-occurrence-free subset contains only 2 problems, too few to inform the interpretation. Potential contributors include the retained reasoning context, upstream answer occurrences, general long-context distraction, interactions among mechanisms, and other factors. On the same 220 sets, the bridge check gives FULL-structured minus original-prompt AGGREGATE $= +0.0227$ $[-0.0182, +0.0682]$. The confidence interval is not contained within the pre-fixed equivalence margin of ± 0.05 , so equivalence was not established (see §3.3).

N Pipeline-rendered timing and resource accounting

Table 3 reports the registered pipeline-rendered, cold-prefix, batch-size-1 timing sample; Table 4 reports token/resource accounting from locked artifacts. The two tables use different samples (the frozen 30-set timing sample versus 220-set full-grid token means) and are therefore reported separately. The timing result characterizes marginal, post-pool generation under the tested execution. Cached-serving and end-to-end latency are outside these measurements, as is candidate-generation cost.

Table 3: Pipeline-rendered, cold-prefix, batch-size-1 marginal request latency on the frozen 30-set timing sample (10 sets per stratum, three unseeded repeats; $n = 30$ observations per cell; prefix caching disabled; matched strategy pairs on the same A100). Latency columns report the median and the empirical p95 of synchronized per-request latency in seconds; token columns are median realized tokens per request; cap hits count outputs at the 16,384-token limit. Descriptive marginal post-pool measurements under the registered configuration — not cached-serving and not end-to-end workflow costs (candidate generation is excluded). This 30-set sample is distinct from the 220-set full-grid token accounting in Table 4.

stratum	strategy	n	med. lat. (s)	emp. p95 lat. (s)	med. in tok	med. out tok	cap hits
c0	AGGREGATE-4	30	13.9	25.8	21,638.5	1,229.5	0
c0	FRESH-SOLVE	30	55.8	101.0	149.5	6,474.0	0
c1	AGGREGATE-4	30	13.0	31.8	21,320.5	1,243.0	0
c1	FRESH-SOLVE	30	59.4	90.2	149.5	6,856.5	1
c2+	AGGREGATE-4	30	14.3	23.5	17,576.0	1,437.5	0
c2+	FRESH-SOLVE	30	38.8	148.7	153.0	4,565.0	2

Table 4: Resource accounting at c0 from locked artifacts. View A: marginal cost after a 4-candidate pool exists (candidate-generation excluded). Token values are *mean realized tokens per request* over 220 c0 sets under the certified chat-templated pipeline; token workloads do not equal compute, and realized lengths differ by design. This table is token/resource accounting only: Task 3b latency values are reported exclusively in the separate 30-set timing-sample table (Table 3); the two samples never share a table row. View B (full workflow incl. candidate generation) is omitted: no committed candidate-generation request identities or token/timing logs exist, so candidate-generation and end-to-end costs are not measured (structurally: candidate generations/samples = 4; post-pool generation calls = 0 or 1; request count depends on batching and is not measured). GPU-seconds and FLOPs: not measured. Token means are full-grid statistics; Task 3b latency values are 30-set sample statistics (medians and empirical p95 over the frozen timing sample) and must carry their own sample/statistic label if ever shown alongside.

strategy (marginal, after pool)	post-pool gen calls	mean in tok	mean out tok
VOTE-4 (CPU plurality)	0	0	0
AGGREGATE-4	1	18,721.7	1,737.2
FRESH-SOLVE (after pool)	1	217.9	8,310.4

O Raw-format timing diagnostic

We preregistered a batch-size-1 latency probe using the original user-prompt text. A post-run consistency check showed that its timing harness omitted the system message and chat template used by the main experimental pipeline. The resulting termination behavior differed substantially: on the same sampled AGGREGATE sets, templated pipeline generations had median output length 1,366 tokens with no truncations, whereas raw-format timing requests had median output length 16,384 tokens and 27 of 30 reached the cap at c0. These measurements accordingly serve as an implementation diagnostic. Latency reporting for the main pipeline comes from the registered pipeline-rendered, cold-prefix study (Appendix N). The raw numerical table remains in the audit archive and supplementary provenance record.

P The reasoning-trained variant

The reasoning-trained model produces a correct extractable answer far more often (CGT 0.936 vs 0.379; Table 7), so 200 of its 240 sets fall in c2+, leaving c0 and c1 with three and two problems. The low- c strata are too sparse to determine whether reasoning training changes the matching cost; the c0 sign is preserved, but its 3-problem stratum is unpowered (Table 7). Within the populated c2+ stratum, the effects have the same direction at smaller magnitude

($\Delta_{\text{cand}} + 0.120$; $\Delta_{\text{reas}} + 0.045$). We treat this result as **exploratory**, with model-class invariance unresolved. NO-CANDIDATE and ANSWER-ONLY carry no-answer truncations that AGGREGATE lacks, so the c2+ Δ_{cand} remains an *upper bound* pending truncation-robust recomputation (truncated-no-answer counts: NO-CANDIDATE 37/240, ANSWER-ONLY 13/240, AGGREGATE 0; Table 9). A no-answer truncation scores wrong, deflating those arms and biasing AGG–NoCand and AGG–AnsOnly upward; the reported c2+ effects are therefore upper bounds under the current scoring.

Q Full result tables

Reading the statistics. All primary tests are *paired sign tests* at the problem level: each problem contributes one mean per arm per c -bin, and we count, across problems, how many favor one arm versus the other. A notation like “10/4” means the first arm was higher on 10 problems and the second on 4 (ties are dropped), and the reported p is the two-sided exact sign-test p -value on those discordant counts; bold in the main table means the entry remains significant after Holm correction within its Δ -family of three bins. Accuracies are fractions correct under a single audited first-answer extractor.

Table 5: Per-benchmark accuracy by arm and c -bin (set-level point estimates). Both benchmarks show the three-regime sign pattern. The AIME $\leftrightarrow$ HMMT homogeneity p (permutation on per-problem Δ_{cand}) is c0 0.91, c1 0.55, c2+ 0.70. Heterogeneity was not detected, and the benchmark-specific effects agree in sign. The pre-specified pooled estimate is reported alongside these per-benchmark results.

bench	c	AGG	NoCand	AnsOnly	Vote	Oracle	Δ_{cand}	Δ_{reas}
AIME	c0	0.222	0.352	0.296	0.000	0.000	−0.130	−0.074
AIME	c1	0.697	0.500	0.618	0.237	1.000	+0.197	+0.079
AIME	c2+	0.893	0.583	0.762	0.976	1.000	+0.310	+0.131
HMMT	c0	0.089	0.205	0.080	0.000	0.000	−0.116	+0.009
HMMT	c1	0.635	0.327	0.423	0.038	1.000	+0.308	+0.212
HMMT	c2+	0.875	0.625	0.600	0.925	1.000	+0.250	+0.275

Table 6: K -sweep (AIME-2025 only; problem-level). Raw AGGREGATE accuracy at c0 decreases as K grows, and Δ_{cand} stays negative at c0 for every K (sign-invariant).

K	raw AGG c0	Δ_{cand} c0	Δ_{cand} c1	Δ_{cand} c2+
2	0.250	−0.170 (.092)	+0.250 (.118)	+0.333 (.007)
4	0.222	−0.130 (.109)	+0.197 (.302)	+0.310 (.022)
8	0.037	−0.188 (.031)	+0.071 (1.00)	+0.274 (< .001)

Table 7: RLVR (Qwen3-4B-Thinking) by c -bin. 200 of 240 sets fall in c2+ (25 of 30 problems); c0/c1 contain 3/2 problems, leaving insufficient power to assess model-class differences. CGT (correct given a valid terminated answer): instruct 244/644=0.379 vs RLVR 395/422=0.936.

c	sets	prob	AGG	NoCand	AnsOnly	Vote	Oracle	Δ_{cand} (p)	Δ_{reas} (p)
c0	24	3	0.083	0.125	0.208	0.000	0.000	−0.042 (1.00)	−0.125 (1.00)
c1	16	2	0.375	0.438	0.500	0.000	1.000	−0.062 (1.00)	−0.125 (1.00)
c2+	200	25	0.960	0.840	0.915	0.980	1.000	+0.120 (.012)	+0.045 (.125)

Table 8: Validation experiment (constructed additive recombination, all c0). The pre-specified success criterion requires AGG to exceed both candidate-free arms by ≥ 0.30 .

quantity	value
AGG (c0)	0.767
NoCand (c0)	0.000
AnsOnly (c0)	0.000
validation margin: AGG – max(NoCand, AnsOnly)	0.767
criterion (≥ 0.30)	PASS

Table 9: Per-arm run health. trunc_noans = finished without an extractable answer; cand_trunc = starved candidate pool. Instruct/HMMT clean (all arms 0). The RLVR counts occur entirely in the non-aggregate arms (NO-CANDIDATE 37, ANSWER-ONLY 13 of 240). Because a no-answer truncation scores wrong, these counts deflate the affected arms and bias AGG–NoCand and AGG–AnsOnly *upward*. The reported RLVR c2+ $\Delta_{\text{cand}} / \Delta_{\text{reas}}$ are therefore upper bounds; the instruct headline is unaffected.

run	arm	n	trunc_noans	cand_trunc
instruct-AIME	AGG/NoCand/AnsOnly	268	0	0
instruct-HMMT	AGG/NoCand/AnsOnly	204	0	0
RLVR	AGGREGATE	240	0	0
RLVR	NO-CANDIDATE	240	37	0
RLVR	ANSWER-ONLY	240	13	0

Table 10: c0 by pool difficulty (pass@24). The c0 deficit is confined to otherwise-solvable problems, while every arm remains at or near zero on unsolvable problems.

bench	stratum	prob	sets	AGG	NoCand	AnsOnly	Δ_{cand}
AIME	pass@24=0	9	36	0.000	0.000	0.000	+0.00
AIME	pass@24 $\geq$ 1	18	72	0.333	0.528	0.444	−0.194
HMMT	pass@24=0	15	60	0.100	0.067	0.000	+0.033
HMMT	pass@24 $\geq$ 1	13	52	0.077	0.365	0.173	−0.288